Exam 2 (November 9)

QTM 220

Problem 1

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. X can take the values of 1,2,3,4 or 5. We run this code.

```
B=10000
theta<- rep(NA,B)
for( b in 1:B ){
bootind = sample(1:n,size=n,replace=T)
dfboot = df[bootind,]
theta[b] = (1/4)*(mean(dfboot$y[dfboot$x==5]) - mean(dfboot$y[dfboot$x==1]))
}
```

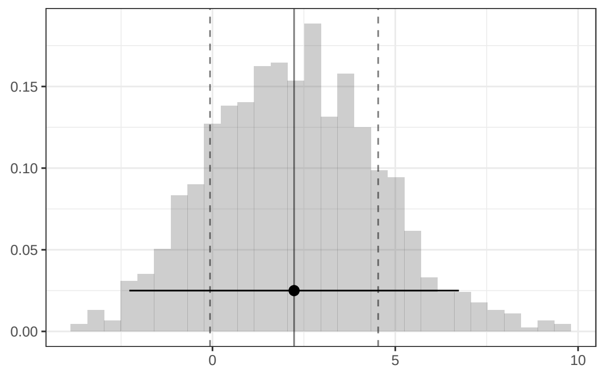


Figure 1: Histogram of the vector Y's elements

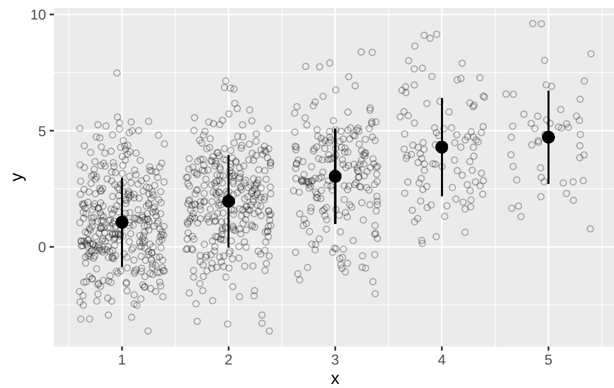


Figure 2: Jittered Scatterplot of X and Y

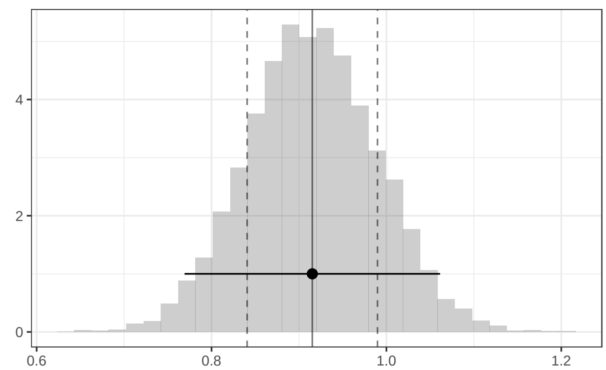


Figure 3: Histogram of the theta's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

Part c

What does this code calculate? State it in words and mark it with a 'c', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y[X==1])
```

Part d

What does this code calculate? State it in words and mark it with a 'd', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(theta)
```

Part e

What does this code calculate? State it in words and mark it with a 'e', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))
- 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))

(1/4)(mean(Y[X=5]) - mean(Y[X=1]))
+ 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))
```

Part f

How could you calculate roughly the same thing as in part e using 'sd(theta)' replacing

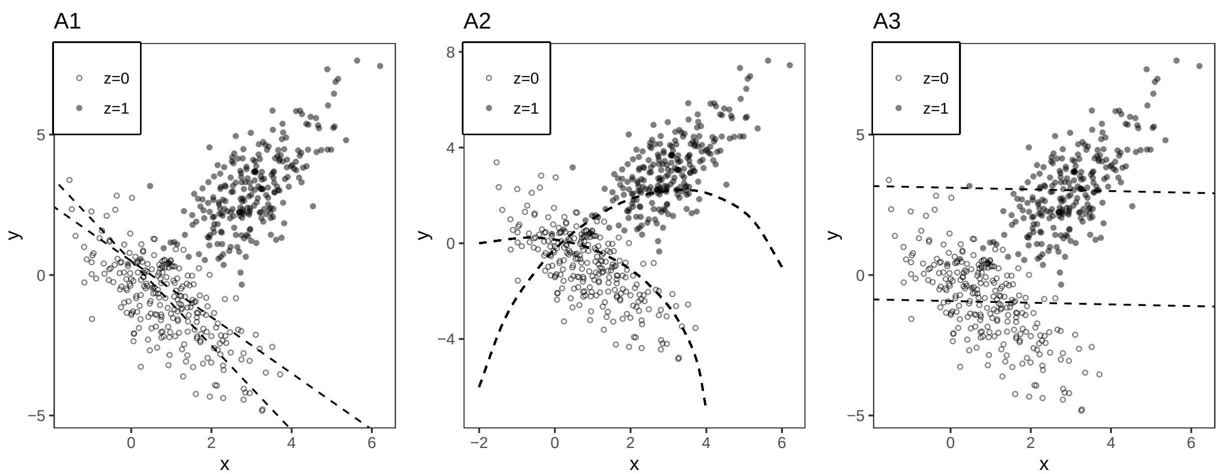
```
sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))
```

Problem 2

In the following parts, choose the figure with the appropriate regression line(s). Clearly mark one choice, no explanation needed.

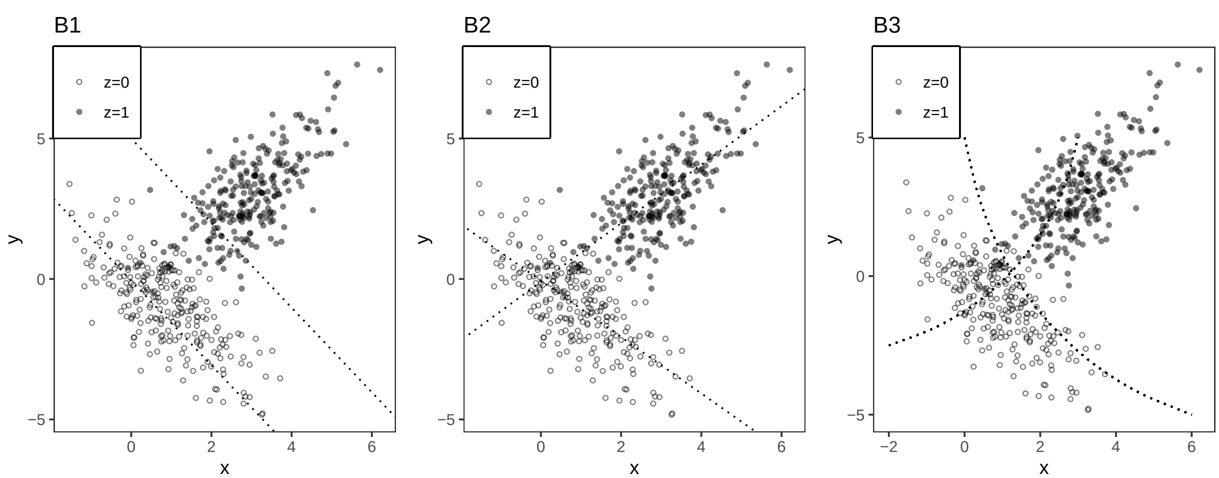
Part A

Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



Part B

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



Problem 3

Part A

We have collected y, x, and a binary variable z. The scatter plots below show our data.

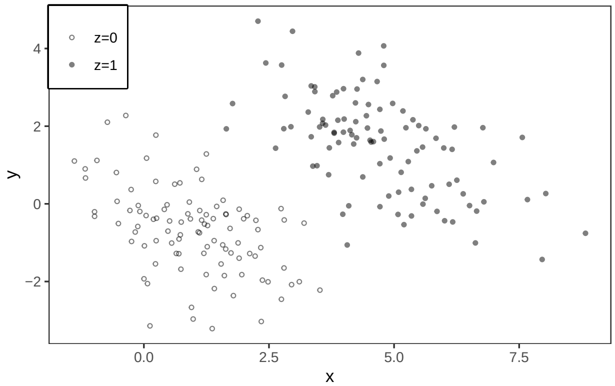


Figure 4: Scatterplot 1

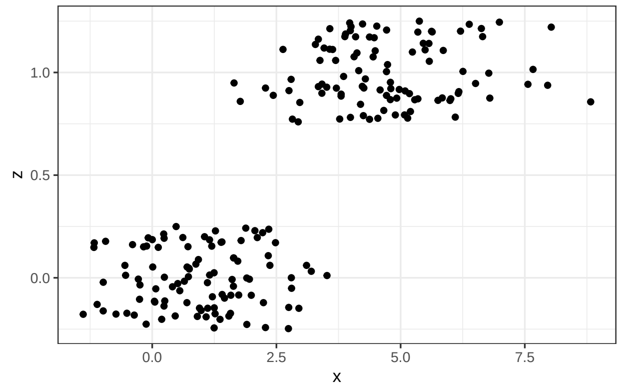


Figure 5: Scatterplot 2 (jittered)

Part A

Do you expect the coefficient on x from the simple regression of y on x ($lm(y \sim x)$) to be: (choose one, no explanation needed)

- Positive
- Negative
- Zero

Part B

We then run the regression $lm(y \sim x + z)$ which gives us the following output:

```
Call:
lm(formula = y ~ x + z, data = dfp3)

Residuals:
    Min      1Q  Median      3Q     Max
-2.9982 -0.6313 -0.0448  0.7168  3.2627

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.07683    0.11938  -0.644    0.521
x           -0.55172    0.05963  -9.253   <2e-16 ***
z            4.21009    0.26966  15.613   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.05 on 197 degrees of freedom
Multiple R-squared:  0.5961,    Adjusted R-squared:  0.592
F-statistic: 145.4 on 2 and 197 DF,  p-value: < 2.2e-16
```

And we run the regression $lm(z \sim x)$ which gives us the following output:

```
Call:
lm(formula = z ~ x, data = dfp3)

Residuals:
     Min       1Q   Median       3Q      Max
-0.62555 -0.20421 -0.00143  0.20566  0.71973

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) -0.023784   0.031416  -0.757     0.45
x            0.184544   0.008657  21.317   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2768 on 198 degrees of freedom
Multiple R-squared:  0.6965,    Adjusted R-squared:  0.695
F-statistic: 454.4 on 1 and 198 DF,  p-value: < 2.2e-16
```

How would you use these outputs to calculate the coefficient on x from the simple regression $lm(y \sim x)$? Explain your steps and define any notation you use. Plug in the estimates from your output (you can leave it as an equation).

Part C

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$ but we run the regression $lm(y \sim x)$ without z or w. We have de-meaned our variables x_i, z_i, and w_i such that $\bar{x} = \sum_{i=1}^{n} x_i = 0$, $\bar{z} = \sum_{i=1}^{n} z_i = 0$, and $\bar{w} = \sum_{i=1}^{n} w_i = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$, be equal to β_1? Derive what your coefficient on x will be in terms of βs and justify your steps.

Problem 4

Consider the following `R` code:

```
1. lm(y ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(x~z)))
3. lm(residuals(lm(y~z)) ~ x)
4. lm(residuals(lm(y~x)) ~ z)
5. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
6. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
7. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
8. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
```

a) Choose one of the above that will produce the coefficient on x from the multiple regression of $y \sim x + z$

b) BONUS: Choose one of the above that will produce the coefficient on z from the multiple regression of $y \sim x + z$

c) Choose one of the above that will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$